



AstraX EB1 Supply Chain War-Room

Critical Components Acceleration & Execution Control

A focused, time-boxed operation to secure every critical, long-lead, and high-cost component required to deliver AstraX EB1 on schedule — with discipline, speed, and zero ambiguity.

Mission Control Objective

Objective

Secure every critical, long-lead, and high-cost component within a 5-day execution window. No gaps, no assumptions — only locked sources and placed orders.

Timeline

Execution window is **immediate: 0-5 days**. Every hour of delay compounds program risk on the EB1 schedule.

Guiding Principle

Speed > Perfection. A good decision made now is worth more than a perfect decision made too late.

Constraint

EB1 program timeline is non-negotiable. Capital efficiency must be preserved — but never at the cost of schedule.

"We are not buying parts. We are securing execution."



Hardware Success = Supply Chain Success

The AstraX EB1 program will not be won in the design lab — it will be won or lost in the supply chain. Every risk below is a direct threat to program delivery.



Long Lead Times

Critical components with 6–16 week lead times must be ordered now to avoid blocking build.



MOQ Traps

Minimum order quantities from distributors can lock capital and create excess inventory if not negotiated.



Obsolescence

EOL and NRND parts threaten downstream builds. Last-time-buy windows close without warning.



Vendor Dependency

Single-source components with no validated alternative create catastrophic program exposure.

Supply Chain Execution Flow

AstraX EB1 procurement runs on a **parallel execution model** — not sequential. Multiple workstreams advance simultaneously to compress total cycle time and surface decision points early.



Key principle: Classification and Sourcing run concurrently with BOM Scrub. Procurement decisions are made at the earliest viable decision gate — waiting for complete information is not an option.

Critical Component Classification

Every part in the EB1 BOM is evaluated against five risk dimensions. Classification drives procurement urgency, sourcing strategy, and executive escalation thresholds.

Flag	Category	Definition & Action Trigger
	Critical	System-blocking component. Absence halts build. Immediate PO required.
	Long Lead	Lead time 6-8+ weeks. Must be ordered before design freeze. No exceptions.
	High Cost	Budget-sensitive item. Requires program manager approval and finance visibility.
	Obsolescence Risk	EOL or NRND status. Triggers last-time-buy evaluation and redesign review.
	Alt Required	No validated backup source exists. Immediate secondary supplier identification required.



TOP 20 WATCHLIST

The Parts That Define the Program

If any of these components slip — through stock-out, lead-time extension, or sourcing failure — **the program fails**. These are the 20 highest-priority parts under active daily tracking.



Imaging Core

- X-ray tube
- Flat-panel detector



Compute & Control

- SOM (i.MX8+)
- MCU (STM32 family)



Power & Regulation

- High-voltage power modules
- DC-DC converters



Sensing & I/O

- Dose sensors
- Position & safety sensors



Any single item in this list without a confirmed source and placed PO is an open program risk. Owners are accountable for daily status updates.

Sourcing Strategy Framework

For every part on the critical list, sourcing is evaluated across four dimensions before a procurement channel decision is made. No part moves to PO without completing this framework.

1

Supplier Mapping

Identify primary and secondary suppliers. Validate authenticity, stock levels, lead time, and MOQ for each.

2

Cost Benchmarking

Obtain minimum three quotes per critical part. Flag deviation >15% for review before approval.

3

Channel Decision

Select the optimal procurement channel based on lead time, cost, and risk profile.

Channel Decision Logic



Authorized Distributor –
Default first choice.
Known stock, traceable.



Direct from Manufacturer – For
high-cost or long-lead
strategic parts.



EMS Partner – Leverage
volume agreements for
cost leverage.



Broker (Last Resort) –
Only with full
authentication protocol.

Risk Mitigation Engine

Every identified supply chain risk maps directly to a mandatory action. No risk is acceptable without an assigned owner and a defined mitigation in progress.

Risk Condition	Severity	Required Action
No second source identified	 Critical	Initiate immediate redesign or alternate component qualification
Lead time >10 weeks	 Critical	Trigger parallel sourcing and explore broker channels with auth protocol
High MOQ exceeds build plan	 High	Negotiate split orders, consignment, or EMS volume pooling
Part confirmed EOL / NRND	 High	Initiate last-time-buy evaluation and expedited redesign review
Single distributor dependency	 Medium	Qualify second authorized distributor within 72 hours



All  Critical risks escalate to program leadership within 24 hours of identification. No exceptions.

Procurement Strategy: Bias for Action

The EB1 procurement posture is **aggressive and decisive**. Analysis paralysis is a program risk. Every procurement decision follows four operational mandates.

1

Immediate PO on Long-Lead Parts

Any part with lead time >6 weeks receives a purchase order before all other actions. No quote process, no approval chain delay — authorize and place.

2

MOQ Optimization

Negotiate split shipments, blanket orders, or EMS pool buying to avoid over-committing capital. Target 1.5× build plan — no more.

3

Last-Time-Buy Decisions

EOL and NRND parts require a LTB quantity decision within 48 hours of identification. Delay = lost supply window.

4

Eliminate Analysis Paralysis

If a decision can be made with 70% confidence, make it now. Reversible decisions are not worth perfect information. Move fast, document rationale.

Build Readiness: Downstream Checklist

Securing components is necessary but not sufficient. The full build chain must be ready to receive and process parts the moment they arrive. Each workstream below has a binary readiness gate.

1

PCB Fabrication

Fab partner confirmed, Gerbers released, NPI slot reserved. No last-minute file revisions.

2

Assembly Partner

EMS partner briefed on EB1 BOM, assembly notes issued, capacity slot held for target build window.

1

Firmware Readiness

Board-bring-up firmware frozen and version-controlled. Dev team aligned to hardware BOM revision.

2

EEPROM & Serial Tracking

Serialization scheme defined, EEPROM programming scripts validated, traceability database initialized.

3

Test Infrastructure

Test jigs designed, fabricated, and validated before first board arrives at test station.

3

Inventory Management

Receiving location confirmed, incoming inspection process defined, kitting instructions issued to floor.

War-Room Execution Dashboard

Every critical part is tracked daily with a RAG status, assigned owner, and next action. This dashboard is the single source of truth for EB1 supply chain health. Red items are discussed first — every day.

Part / Component	Risk	Status	Owner	Next Action
SOM (i.MX8+)	● Critical	Pending Quote	Procurement Lead	Request quotes from 3 distributors by EOD
X-Ray Tube	● High	Quote Received	Sourcing Manager	PO approval — PM sign-off required
Flat-Panel Detector	● Critical	Single Source	Engineering Lead	Initiate alt-source qualification immediately
HV Power Module	● High	PO Placed	Procurement Lead	Confirm delivery date with supplier
MCU (STM32)	● Medium	Stock Confirmed	Ops Team	Release PO against confirmed stock

 Dashboard is updated every morning at 9 AM. Owners are responsible for updating their rows before stand-up. Stale data is treated as a red flag.

The 5-Day Execution Clock

This is not a roadmap — it is a commitment. Every milestone below is mandatory. Misses escalate to program leadership within the same business day.

1

🕒 24 Hours

Classified BOM Complete. Every part assigned a risk flag. Owners identified. No unclassified parts.

2

🕒 72 Hours

Sourcing Locked. Primary and secondary sources confirmed for all critical parts. Quotes in hand.

3

🕒 5 Days

All POs Placed. Every critical and long-lead part has an issued purchase order with a confirmed delivery date.



Teams that execute this window on time give AstraX EB1 its best possible chance of on-schedule delivery. The clock is running.



"AstraX success will be decided in supply chain execution – not design intent."

The design is ready. The intent is clear. Now the work begins — in sourcing channels, purchase orders, and daily stand-ups. **Execute without hesitation.**

Own Your Parts

Every critical component has one owner. That owner is accountable — full stop.

Escalate Fast

Surface blockers immediately. Hidden problems become program delays. Transparency is non-negotiable.

Move Now

The 5-day window has started. Every decision delayed is a risk added. Speed is the strategy.

